

Evaluation report: six-month platform

Six-month roadmap completed in a 2D digital twin (Gazebo/Isaac interchangeable later).

True open aisle success	0.9 adversarial failure rate	0.3 random failure rate	19.93 tuning improvement pct	6 grammar maps	True cbf open success
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Experiment	Passed	Notes
adversarial_failure_search	yes	Adversarial/MCMC find a higher failure rate than uniform random.
parameter_optimization	yes	Bayesian search matches or beats grid with fewer or equal evals, and stays collision-free at best speed.
sim_to_sim_transfer	yes	RL policy is trained under domain randomization and evaluated at three friction values (Gazebo vs Isaac proxy).
sensor_degradation	yes	Policy trained with randomized sensor noise is evaluated under dropout; baseline P-control is compared.
grammar_scenarios	yes	
safety_verification	yes	

This evaluation runs entirely on the in-repo 2D digital twin. Physics engines such as Gazebo or Isaac Sim can replace the virtual lab later; the adversarial search, Bayesian tuning, CBF shield, domain-randomized RL, and grammar layouts stay the same.